

## Honors Calculus III, First midterm

This exam consists of two problems for a total of 30 points. Attempt to solve as many problems as you can. Make sure you read each question carefully and understand what you are being asked to do. You should progress sequentially, justify each step, use notation correctly, and clearly mark the final answer. Please write your name and section number on every answer booklet you use.

**In this exam you can use results from lectures or homework, provided you state them clearly and you are not explicitly asked to prove them in the question. As usual, for full credit you need to show and explain your work.**

**Problem 1.** Let  $f_n : \mathbf{R} \rightarrow \mathbf{R}$  be a sequence of functions, and assume that  $f_n$  converges pointwise to  $f : \mathbf{R} \rightarrow \mathbf{R}$ .

1. (7pt) Prove that  $f_n$  converges to  $f$  uniformly on  $\mathbf{R}$  if and only if  $\lim_{n \rightarrow \infty} \sup_{x \in \mathbf{R}} |f_n(x) - f(x)| = 0$ .
2. (7pt) Prove that  $f_n$  converges to  $f$  uniformly on  $\mathbf{R}$  if and only if the following is true: for all  $\epsilon > 0$ , there exists  $N \geq 0$  such that  $n, m \geq N$  implies that  $|f_n(x) - f_m(x)| < \epsilon$  for all  $x \in \mathbf{R}$ .
3. (6pt) Let

$$g_n(x) = \frac{x^2 + 1}{e^x + n}.$$

Determine with proof whether the sequence  $g_n$  converges uniformly on  $[0, 1]$ .

**Problem 2.**

1. (3pt) Let  $F(x) = \sum_{n \geq 1} \frac{x^n}{n^2}$ . Prove that the power series

$$\sum_{n \geq 1} \frac{x^n}{n^2}$$

converges for all  $x \in (-1, 1)$ .

2. (4pt) Let  $f(x) = \sum_{n \geq 1} \frac{x^n}{n^2}$  for all  $x \in (-1, 1)$ . Prove that  $f : (-1, 1) \rightarrow \mathbf{R}$  is differentiable, and that

$$f'(x) = -\frac{\log(1-x)}{x}$$

for all  $x \in (-1, 1), x \neq 0$ .

3. (3pt) Let  $x_1, \dots, x_n \in \mathbf{R}$  be real numbers. Prove that

$$\frac{(x_1 + x_2 + \dots + x_n)^2}{n} \leq x_1^2 + x_2^2 + \dots + x_n^2.$$

### Solutions to Problem 1.

1. This was done in class.
2. Assume that  $f_n \rightarrow f$  uniformly on  $\mathbf{R}$ , and fix  $\epsilon > 0$ . Then there exists  $N > 0$  such that  $n \geq N$  implies  $|f_n(x) - f(x)| < \epsilon/2$  for all  $x \in \mathbf{R}$ . Hence if  $n, m \geq N$  then

$$|f_n(x) - f_m(x)| \leq |f_n(x) - f(x)| + |f_m(x) - f(x)| < \epsilon/2 + \epsilon/2 = \epsilon.$$

On the other hand, assume that the given statement is true and fix  $\epsilon > 0$ . Let  $N > 0$  be such that  $n, m \geq N$  implies  $|f_n(x) - f_m(x)| < \epsilon/2$  for all  $x \in \mathbf{R}$ . Now fix  $n \geq N$  and  $x \in \mathbf{R}$ , and let  $m \geq N$  tend to infinity: we find that

$$|f_n(x) - f(x)| = \lim_{m \rightarrow \infty} |f_n(x) - f_m(x)| \leq \epsilon/2 < \epsilon.$$

So we have just shown that  $n \geq N$  implies  $|f_n(x) - f(x)| < \epsilon$  for all  $x \in \mathbf{R}$ . This means that  $f_n$  converges uniformly to  $f$  on  $\mathbf{R}$ .

3. If  $x \in [0, 1]$  is fixed we find that  $\lim_{n \rightarrow \infty} f_n(x) = 0$ , hence the pointwise limit is zero. To show uniform convergence we will use the criterion from part (1). In fact, we have

$$|f_n(x)| \leq \frac{2}{n}$$

for all  $x \in [0, 1]$ , since if  $x \in [0, 1]$  then  $x^2 + 1 \leq 2$  and  $e^x + n \geq n$ . Hence

$$0 \leq \sup_{x \in [0, 1]} |f_n(x)| \leq \frac{2}{n} \rightarrow 0 \text{ as } n \rightarrow \infty,$$

and so we are done by part (1).

### Solutions to Problem 2.

1. If  $x \in (-1, 1)$  then  $|x| < 1$ , and so the given power series converges absolutely by the comparison test, since  $\sum_{n \geq 1} \frac{1}{n^2}$  converges.
2. By definition, part (1) implies that the radius of convergence  $R$  of  $\sum_{n \geq 1} \frac{x^n}{n^2}$  is greater than or equal to 1: if  $R < 1$  then the power series would not converge for  $x \in (R, 1)$ . By a theorem from class, this implies that  $f(x)$  is a differentiable function on  $(-1, 1)$ , and

$$f'(x) = \sum_{n \geq 1} \frac{x^{n-1}}{n}$$

for all  $x \in (-1, 1)$ . It follows that

$$xf'(x) = \sum_{n \geq 1} \frac{x^n}{n}.$$

On the other hand, we know that

$$\log(1+x) = \sum_{n \geq 1} (-1)^{n+1} \frac{x^n}{n}$$

for all  $x \in (-1, 1)$ . Since  $x \in (-1, 1)$  if and only if  $-x \in (-1, 1)$ , this implies that

$$-\log(1-x) = -\sum_{n \geq 1} (-1)^{2n+1} \frac{x^n}{n} = \sum_{n \geq 1} \frac{x^n}{n}$$

for all  $x \in (-1, 1)$ . Hence  $xf'(x) = -\log(1-x)$  for all  $x \in (-1, 1)$ , since these functions are given by the sum of the same infinite series. This concludes the proof since we can divide by  $x$  if  $x \neq 0$ .

3. We use the Cauchy–Schwarz inequality for the vectors

$$x = (x_1, x_2, \dots, x_n) \in \mathbf{R}^n \text{ and } y = (1, 1, \dots, 1) \in \mathbf{R}^n.$$

We find that

$$|\langle x, y \rangle|^2 \leq \|x\|^2 \|y\|^2$$

and so

$$(x_1 + x_2 + \dots + x_n)^2 \leq (x_1^2 + x_2^2 + \dots + x_n^2)n.$$

Dividing by  $n$  we get the result.